

KIRCHOFF'S CURRENT AND VOLTAGE LAWS (KCL and KVL)

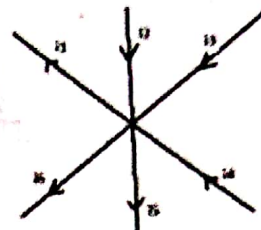
KIRCHOFF'S CURRENT LAW : KCL states that the total current entering a junction is equal to the total current leaving the junction.

(or)

The algebraic sum of the currents at the junction (node) will be zero.

At node n, $(i_2 + i_3 + i_4) = (i_1 + i_6 + i_5)$

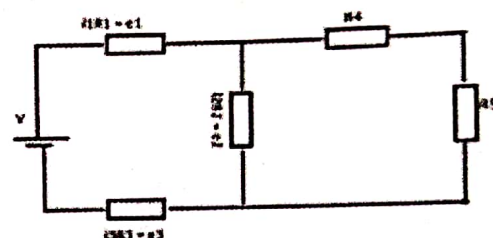
Or $i_2 + i_3 + i_4 - i_1 - i_6 - i_5 = 0.$



KIRCHOFF'S VOLTAGE LAW: KVL is based on the law of conservation of the energy, states that the algebraic sum of voltage drops in a closed loop is zero.

$$e_1 + e_2 + e_3 = V$$

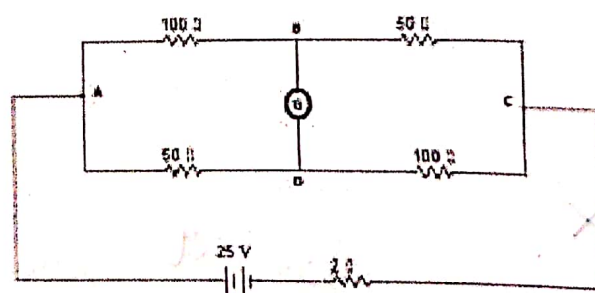
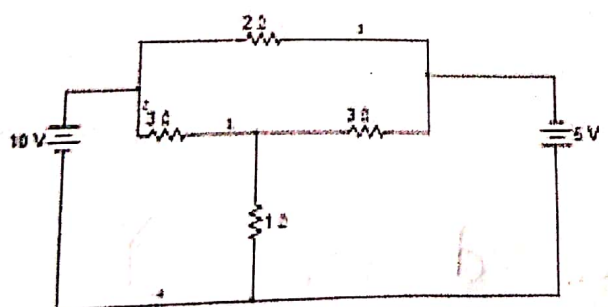
$$e_1 + e_2 + e_3 - V = 0$$



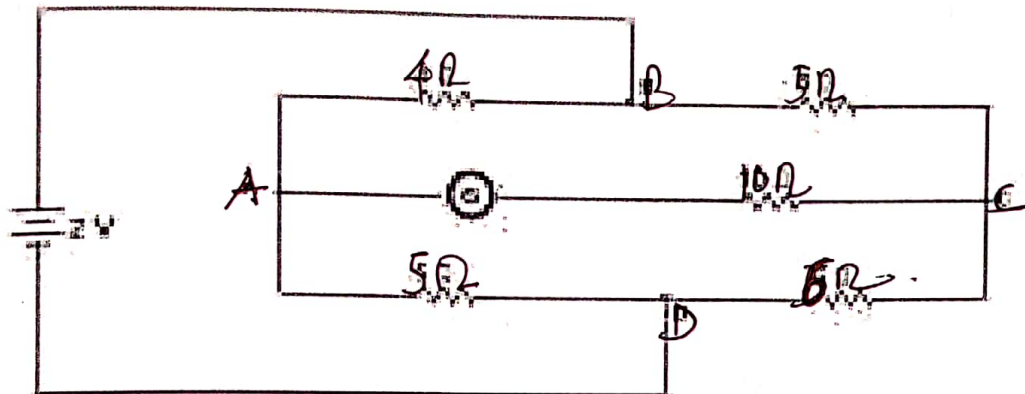
{Flow of currents in loop is assumed +ve from higher potential to lower potential in elements and +ve from lower to higher potential in Sources}

Group 1.

1. Write KCL and KVL equations for the given circuits:

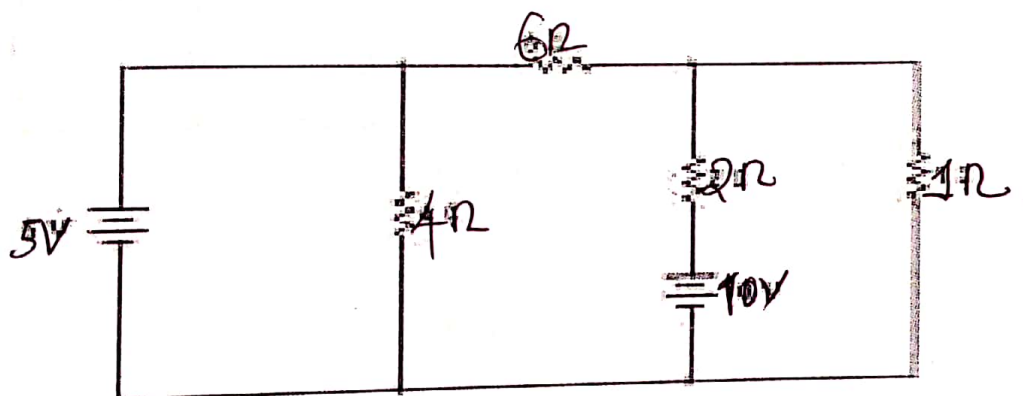


2. Find the current measured in galvanometer shown in the wheatstone's bridge.



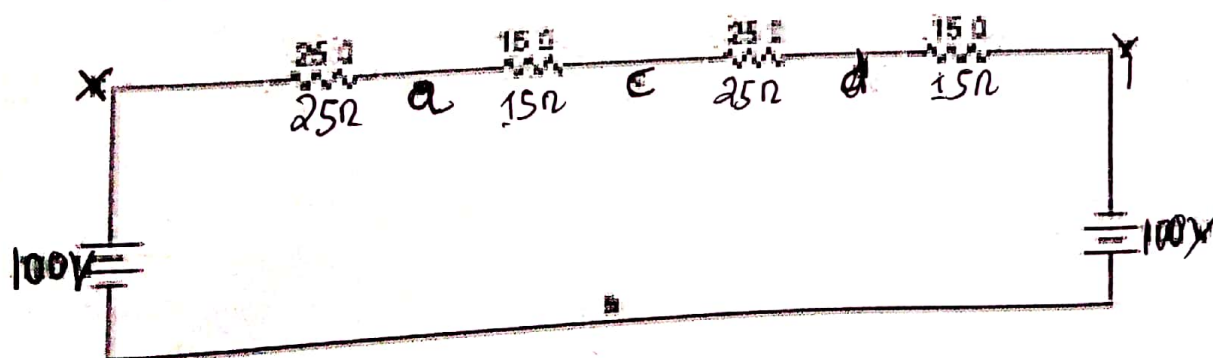
Group 2
(1)

3. Determine the voltage across 1Ω resistor in the given circuit.



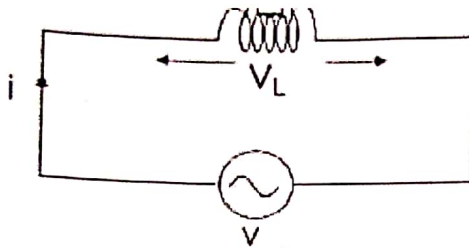
(2)

4. From the given circuit, find the voltage across ab, cb, and db.



AC circuit with a pure inductance

Consider an AC circuit with a pure inductance L as shown in the figure. The alternating voltage v is given by



We can find the current through the inductor as follows

$$i = L \frac{di}{dt}$$

$$v_m \sin(\omega t) = L \frac{di}{dt}$$

$$di = \frac{v_m}{L} * \sin(\omega t)$$

$$i = \frac{v_m}{L} * \int \sin(\omega t) dt$$

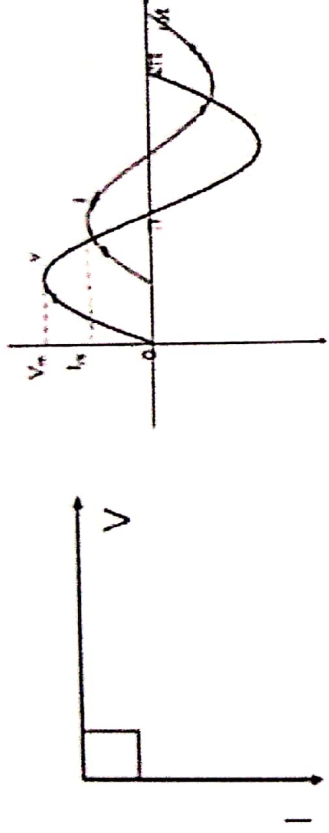
$$i = \frac{v_m}{\omega L} * -\cos(\omega t)$$

$$i = \frac{v_m}{\omega L} * \sin(\omega t - \frac{\pi}{2})$$

$$i = i_m * \sin(\omega t - \frac{\pi}{2})$$

$$\text{Where } i_m = \frac{v_m}{\omega L}$$

From equation (1) and (2) we observe that in a pure inductive circuit, the current lags behind the voltage by 90° . Hence the voltage and current waveforms and phasors can be drawn as below.



Inductive reactance

The inductive reactance X_L is given as

$$X_L = 2\pi fL$$

$$i_m = \frac{v_m}{X_L}$$

in phasor form $\bar{V} = j\omega L\bar{I}$

It is equivalent to resistance in a resistive circuit. The unit is ohms ()

Problem:

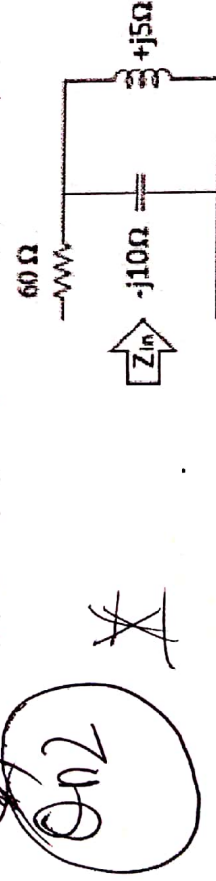
The voltage $v = 12 \cos(60t + 45^\circ)$ is applied to a 0.1H inductor. Find the steady-state current through the inductor.

Problem:

The voltage $v = 10 \cos(100t + 30^\circ)$ is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor, and Draw phasor diagram?

Problem: Group 4

Find equivalent impedance of below circuit.

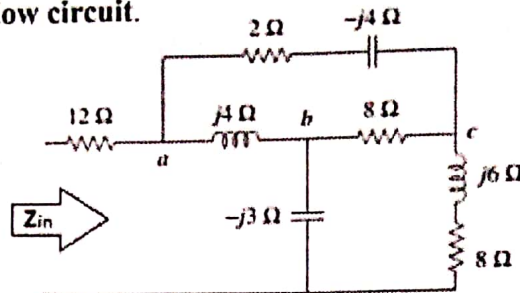


Group 3

Problem:

Qn 1

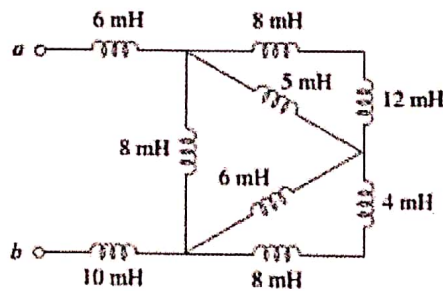
Find impedance Z_{in} of below circuit.



Problem:

Qn 2

Find L_{eq} between 'ab' terminals in below figure?

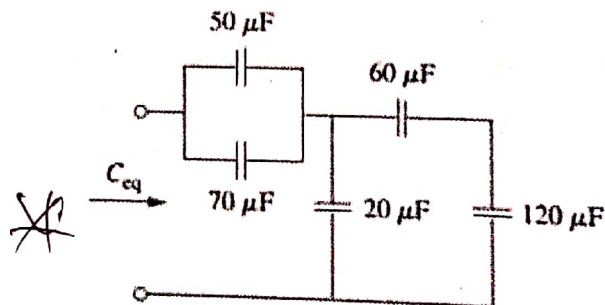


Group 4

Qn 1

Problem:

Find C_{eq} in below figure.



Group 5

Explain Thevenin's Theorem with a suitable example.

~~For Qn 1~~

Find V_{TH} , R_{TH} and the load current flowing through and load voltage across the load resistor in fig (1) by using Thevenin's Theorem

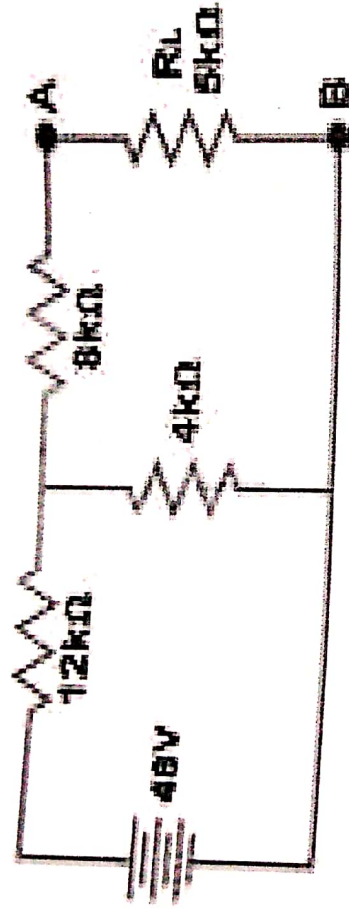


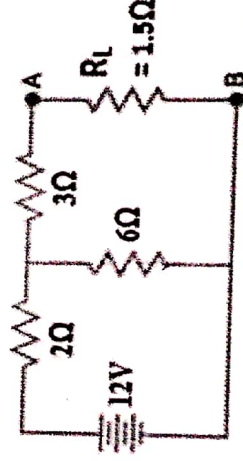
Fig 1

Group 6

Q4) Explain Norton's Theorem with suitable example.

Example:

Find R_N , I_N , the current flowing through and Load Voltage across the load resistor in fig (1) by using Norton's Theorem.



Group 7

Q5) Explain Superposition Theorem with a suitable example.

Q6) Explain Maximum Power Transfer Theorem and derive an expression for P_{max}